

Synopsis Report

CareSlot – Appointment Management System

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Project Title

CareSlot – Appointment Management System

1. Introduction

The **CareSlot Appointment Management System** is a front-end web application designed to simplify and organize appointment booking at a small clinic or hospital reception. Traditional appointment handling often relies on paper registers and phone slips, which are time-consuming, easy to lose, and prone to mistakes such as two patients being booked with the same doctor at the same time. This project provides a single-page digital platform where reception staff can book, track, and manage patient appointments through a clean and user-friendly website.

The project is developed using **HTML5**, **CSS3**, **JavaScript (ES6)**, and the **Bootstrap 5** framework, focusing on an attractive interface, working logic, and an organized layout. Appointment data is saved in the browser's `localStorage`, so it remains available after the page is refreshed.

2. Problem Statement

Many small clinics still manage appointments manually. Entries in paper registers are hard to search, phone bookings are written on slips that get lost, double bookings are discovered only when patients arrive, and nobody keeps an accurate count of completed or cancelled visits. This leads to delays, wasted time, and confusion for both patients and staff.

3. Proposed Solution

The proposed CareSlot system provides one screen where the user can:

- Book an appointment with patient name, phone number, doctor, date, time, and notes
- Have every entry checked automatically, including past-time and double-booking checks
- View all appointments in a table sorted by date and time
- Search appointments by patient or doctor name
- Filter appointments by status (Upcoming, Completed, Cancelled)
- Edit or reschedule an appointment
- Mark an appointment as completed or cancelled, or delete it
- See live statistics of total, upcoming, completed, and cancelled appointments

This system reduces paperwork, prevents scheduling clashes, and makes appointment management faster and more organized.

4. Objectives

- To digitize the appointment booking process.
- To reduce booking mistakes, especially double booking.
- To give a quick overview of the clinic's daily workload.
- To make it easy to track the outcome of every appointment and find any record quickly.
- To create a clean, responsive, and user-friendly interface.

5. Scope of the Project

The project is suitable for small clinics, hospitals, and health centres where reception staff need a quick and simple way to manage appointments. It is a single-page front-end application with no server, no database, and no login. Data is stored in the browser of the device being used. It serves as a working prototype that can later be integrated with backend technologies and a database to support multiple users and devices.

6. Modules of the Project

Navigation and Hero Section

Displays the brand name, navigation links to the booking form and appointment list, and a welcome headline.

Statistics Dashboard

Shows live counts of total, upcoming, completed, and cancelled appointments, updated instantly after every change.

Booking Form

Allows the user to enter patient name, phone number, doctor or service, date, time, and optional notes to book an appointment.

Validation Module

Checks every entry before saving: valid name, 10-digit mobile number, doctor chosen, date and time present, time not in the past, and no double booking for the same doctor.

Appointment List

Displays all appointments in a table with patient details, doctor, date and time, notes, and a coloured status badge, sorted by date and time.

Search and Filter

Lets the user search by patient or doctor name and filter by appointment status, with results updating as they type.

Edit and Reschedule

Loads an existing appointment back into the form so details can be corrected or the date and time changed.

Status Management and Delete

Allows an upcoming appointment to be marked as completed or cancelled, and any appointment to be deleted after confirmation.

Data Persistence

Saves all appointments in the browser's `localStorage` so they remain after the page is refreshed or the browser is closed.

7. Technologies Used

- HTML5
- CSS3 (custom properties, Flexbox, media queries, animations)
- JavaScript (ES6)
- Bootstrap 5.3.3
- Google Fonts (DM Sans and Fraunces)
- Browser `localStorage`

8. Expected Outcomes

- Fast and error-free appointment booking.
- Prevention of double bookings and past-time bookings.
- A clear daily overview through live statistics.
- Reduced paperwork and lost records.
- Organized appointment information that is easy to search and filter.
- Improved user experience on both mobile and desktop screens.

9. Future Enhancements

- Staff and Doctor Login
- Database Integration with a shared server
- SMS and Email Reminders
- Patient Self-Booking of Free Slots
- Appointment Length and Slot Grid to Prevent Overlapping Times
- Doctor Working Hours and Leave Days
- Patient Records and Visit History
- Doctor Dashboard
- Export to CSV and Printing of Daily Lists
- Online Fee Payment
- Reports and Analytics
- Dark Mode

10. Advantages

- User-friendly interface
- Easy navigation on one page
- Time-saving booking process
- Automatic checks that prevent mistakes
- Organized and searchable information
- Data is kept after refresh
- Practical and scalable
- Beginner-friendly implementation

11. Conclusion

The **CareSlot Appointment Management System** is a simple, efficient, and practical web-based project that demonstrates how digital solutions can improve appointment management in a clinic. By bringing booking, searching, rescheduling, status tracking, and statistics onto a single screen, the system enhances convenience, accuracy, and organization for reception staff. Although currently implemented as a front-end application that stores data in the browser, it provides a strong foundation for future development into a complete clinic management system with backend functionality, multiple users, and automated reminders.